WORKED SOLUTION

STEP I 2013, Question 12

Probability and Statistics • Sampling Without Replacement

STEP I 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q12

The key idea

Choosing k tablets "at random" means every set of k tablets left in the container is equally likely, so each day's probability is a ratio of counts: favourable selections over $\binom{\text{tablets left}}{k}$. For the last-day question, the trick is symmetry: the tablets left for the final day are just as random as the ones picked on the first. In (iii), the days chain together nicely, because after a good day the container looks exactly like a shorter holiday.

Solution

Throughout, think of the tablets as distinguishable (say, numbered), even though they look the same. This doesn't change any probability, and it makes all $\binom{N}{k}$ choices of k tablets from N equally likely.

Part (i)

With $k = 3$ and $n = 3$ there are 9 tablets, 3 of each type. The number of ways to choose 3 of them is $\binom{9}{3} = 84$. To get one of each type, we choose one of the 3 tablets of type 1, one of the 3 of type 2 and one of the 3 of type 3, so there are $3^3 = 27$ ways. Hence

$$\mathrm{P}(\text{one of each on day 1}) = \frac{27}{84} = \frac{9}{28},$$

as required.

Ask yourself Why don't I need to work through what happens on days 1 and 2 to find the last-day probability?

Over the holiday, the nine tablets are split at random into three groups of three: the day-1 group, the day-2 group and the day-3 group (what's left). Choosing the groups in the order 1, 2, 3 or in the order 3, 2, 1 gives the same distribution: each way of splitting the tablets into three labelled groups is equally likely. In particular, the set left for day 3

is a uniformly random set of three tablets, exactly like the set picked on day 1. So the probability that I am left with one tablet of each type on the last day is also

$$\frac{9}{28}.$$

Part (ii)

Now there are $3n$ tablets, n of each type. There are $\binom{3n}{3}$ equally likely choices, and n^3 of them contain one of each type. So

$$\begin{aligned}\mathrm{P}(\text{one of each on day 1}) &= \frac{n^3}{\binom{3n}{3}} = \frac{6n^3}{3n(3n-1)(3n-2)} \\ &= \frac{2n^2}{(3n-1)(3n-2)}.\end{aligned}$$

For $n = 3$ this is $\frac{18}{8\cdot 7} = \frac{9}{28}$, matching (i).

Part (iii)

Now $k = 2$ and there are $2n$ tablets, n of each of the two types. For $m = n, n-1, \ldots, 1$, let G_m be the event that I select one of each type on the day when m tablets of each type are in the container at the start of the day (so $m = n$ on day 1).

If the container holds m of each type, there are $\binom{2m}{2} = m(2m-1)$ choices of two tablets, and $m \cdot m$ of them are one of each type. So

$$\mathrm{P}(\text{one of each} \mid m \text{ of each left}) = \frac{m^2}{m(2m-1)} = \frac{m}{2m-1}.$$

If every day so far has been good, then each day has removed exactly one of each type, so the next day starts with $m-1$ of each. Multiplying the conditional probabilities day by day,

$$\mathrm{P}(\text{one of each type every day}) = \prod_{m=1}^{n} \frac{m}{2m-1} = \frac{n!}{1 \times 3 \times 5 \times \cdots \times (2n-1)}.$$

To tidy the product of odd numbers, multiply top and bottom by the even numbers $2 \times 4 \times \cdots \times (2n) = 2^n\, n!$:

$$1 \times 3 \times \cdots \times (2n-1) = \frac{(2n)!}{2^n\, n!}.$$

Hence

$$\boxed{\mathrm{P}(\text{one of each type every day}) = \frac{2^n\,(n!)^2}{(2n)!}}$$

(As a check, $n = 1$ gives 1, which is right: with one tablet of each type I must take both.)

Now use Stirling's approximation for $n!$ and for $(2n)!$:

$$(n!)^2 \approx 2n\pi \left(\frac{n}{\mathrm{e}}\right)^{2n},$$
$$(2n)! \approx \sqrt{4n\pi}\left(\frac{2n}{\mathrm{e}}\right)^{2n} = 2\sqrt{n\pi}\,2^{2n}\left(\frac{n}{\mathrm{e}}\right)^{2n}.$$

The factors $(n/\mathrm{e})^{2n}$ cancel, leaving

$$\frac{2^n\,(n!)^2}{(2n)!} \approx \frac{2^n \cdot 2n\pi}{2\sqrt{n\pi}\,2^{2n}} = 2^{-n}\,\frac{n\pi}{\sqrt{n\pi}} = 2^{-n}\sqrt{n\pi},$$

which is the required approximation.

Checks

- Exhaustive counting by computer of all ways of drawing the tablets confirms $\frac{9}{28}$ for both day 1 and day 3 in (i), the formula in (ii) for $n \leq 7$, and the formula in (iii) for $n \leq 4$.
- For $n = 10$ the exact probability in (iii) is $\frac{2^{10}(10!)^2}{20!} \approx 0.005542$, while $2^{-10}\sqrt{10\pi} \approx 0.005474$: about 1.3% low. The ratio tends to 1 as n grows (it is about 1.0025 at $n = 50$), as you'd expect from Stirling.
- Alternative count for (iii): split the $2n$ tablets into an ordered list of n pairs in $(2n)!/2^n$ equally likely ways; the good splits pair the type-1 tablets with the type-2 tablets day by day in $n! \times n!$ ways. The ratio is again $2^n(n!)^2/(2n)!$.

Common slips

- Drawing one tablet at a time and writing $\frac{3}{9} \times \frac{3}{8} \times \frac{3}{7}$, which is the chance of types 1, 2, 3 *in that order*. The types can come in 3! orders, giving $6 \times \frac{27}{504} = \frac{9}{28}$. (Or: the first tablet can be anything, so $1 \times \frac{6}{8} \times \frac{3}{7} = \frac{9}{28}$.)
- Trying to condition on everything that happens on day 1 for the last-day probability; it works, but is long, and symmetry gives the answer at once.
- In Stirling for $(2n)!$, replacing n by $2n$ everywhere, including inside the square root: $\sqrt{2(2n)\pi} = 2\sqrt{n\pi}$, not $\sqrt{2n\pi}$.

Beyond STEP

The answer to (iii) is $2^n/\binom{2n}{n}$, and the estimate $\binom{2n}{n} \approx 4^n/\sqrt{n\pi}$ appears all over probability: it gives the chance that a simple random walk is back at its starting point after $2n$ steps, and it drives the proof that such a walk in one or two dimensions returns home with probability 1. The symmetry argument in (i) is the idea of *exchangeability*: the order in which random draws are made doesn't change their joint distribution. Both ideas

are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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